Question	Answer	Marks	Partial Marks
5(c)	Tangent ruled at $x = -2$	**1**	**B1** for correct tangent
	–4.5 to –2.5	**2**	**Dep** on B1 for tangent or close attempt at tangent at $x = -2$ **M1** for rise/run **also dep on** tangent drawn or close attempt at correct tangent Must see correct or implied calculation from a drawn tangent
5(d)(i)	8, 4, 0.25 oe	**3**	**B1** for each
5(d)(ii)	Correct graph	**3**	**B2FT** for 6 or 7 correct plots or **B1FT** for 4 or 5 correct plots
5(d)(iii)	1.8 to 1.9	**1**	
6(a)	40.5 or 40.45[8..] or 40.46 nfww	**4**	**M1** for 25, 32.5, 37.5, 50, 80 soi **M1** for Σft **M1 dep** for their $\Sigma ft \div 120$
6(b)	Fully correct histogram	**4**	**B1** for each correct bar If 0 scored, **SC1** for frequency densities of 5.4, 4.2, 0.8 and 0.45 seen
7(a)	[y =] $4x + 5$	**3**	**B2** for answer [y =] $4x + c$ oe (c can be numeric or algebraic) OR **M2** for $\frac{y-9}{x-1} = \frac{9-(-3)}{1-(-2)}$ oe OR **M1** for $\frac{9--3}{1--2}$ oe or for **M1** for correct substitution of (–2, –3) or (1, 9) into $y = (their\ m)x + c$ oe
7(b)	76[.0] or 75.96...	**2**	**M1** for tan[ ] = 4 oe

Question	Answer	Marks	Partial Marks
7(c)(i)	$[y =]\ -\frac{1}{4}x+\frac{23}{8}$ oe	**3**	**B2FT** for $[y =]\ -\frac{1}{\textit{their}\text{ gradient from }\textbf{(a)}}x+c$ oe (c can be numeric or algebraic) OR **M2** for $\frac{y-2}{x-3.5}=-\frac{1}{\textit{their}\text{ gradient from }\textbf{(a)}}$ oe OR **M1** for $-\frac{1}{\textit{their}\text{ gradient from }\textbf{(a)}}$ soi **M1** for correct substitution of (3.5, 2) into $y = (\textit{their } m)x + c$ oe
7(c)(ii)	(–4.5, 4)	**2**	**B1** for each value or for $\begin{pmatrix}-8\\2\end{pmatrix}$ seen
8(a)(i)	$\frac{x-1}{x+2}$	**2**	**B1** for either numerator or denominator correct
8(a)(ii)(a)	$\frac{x}{x+3}\times\frac{x-1}{x+2}=\frac{7}{15}$	**B1**	**FT** *their* **(a)(i)** = $\frac{7}{15}$
	$15x(x – 1) = 7(x + 3)(x + 2)$	**M1**	Removes all algebraic fractions **FT** *their* equation if in comparable form
	$15x^2 – 15x = 7x^2 + 21x + 14x + 42$	**M1**	Correctly expands all brackets **FT** *their* equation if in comparable form
	$[8x^2 – 50x – 42 = 0]$ $4x^2 – 25x – 21 = 0$	**A1**	With no errors or omissions seen and one further stage seen after final M1
8(a)(ii)(b)	$(4x + 3)(x – 7)$ [= 0]	**M2**	**M1** for $4x(x – 7) + 3(x – 7)$ or $x(4x + 3) – 7(4x + 3)$ or for $(4x + a)(x + b)$ where either $ab = –21$ or $4b + a = –25$ If 0 scored, **SC1** for $4x + 3$ **and** $x – 7$ seen but not in factorised form
	7 and $-\frac{3}{4}$	**B1**	
8(a)(ii)(c)	7	**1**	**FT** *their* positive solution